Bank Data Analysis – ELT Assignment Report

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Comprehensive Bank Data Analysis Report

This report provides a detailed overview of the data analysis project on the world's largest banks by market capitalization, drawing from the ETL-Project-Bank-Data-Analysis-Starter.ipynb notebook.

1. Project Overview

The objective of this project is to use Apache Spark for a structured analysis of large-scale banking data. The analysis aims to uncover key insights, such as market capitalization trends and global rankings, to support financial decision-making. The business value is derived from enabling financial institutions to monitor their market position, assess performance, and optimize their strategies in a competitive global market. The project also includes storing the processed data in AWS S3 for scalability and accessibility.

2. Data Sourcing and Preparation

The dataset was obtained from an archived Wikipedia page on the

world's largest banks as of September 2023. It consists of a single table with three attributes: Rank, Bank Name, and Market Cap (USD Billion). The data was programmatically scraped using pandas from the HTML document and converted into a PySpark DataFrame.

3. Data Cleaning and Preprocessing

A thorough data cleaning process was performed to ensure the data was ready for analysis.

Handling Missing Values: A check for missing values was performed, and it was confirmed that no null records were present in the dataset.

Duplicate Records: The dataset was scanned for duplicates, and the analysis confirmed there were no duplicate rows.

Outlier Analysis: An outlier analysis was conducted on the Market Cap (USD Billion) column using the interquartile range (IQR) method. JPMorgan Chase, as the top-ranked bank, was identified as a statistical outlier due to its significantly higher market capitalization. However, it was intentionally kept in the dataset as it represents a genuine data point crucial for understanding

market leadership.

4. Key Findings from Exploratory Data Analysis

The exploratory data analysis (EDA) revealed significant patterns and insights into the global banking market.

Market Dominance: Visualizations such as a bar chart of the top 10 banks and a pie chart of market share distribution confirm the dominant position of JPMorgan Chase. The cumulative market share analysis shows that the top three banks (JPMorgan Chase, Bank of America, and Industrial and Commercial Bank of China) account for more than 50% of the total market capitalization. This highlights a strong market concentration.

Distribution of Market Capitalization: A histogram of the Market Cap (USD Billion) shows a skewed distribution, with a high frequency of banks in the lower market cap ranges and a sharp drop-off towards the higher end. This emphasizes the significant gap between the top few banks and the rest of the list.

Rank vs. Market Cap Relationship: A scatter plot of Rank versus Market Capitalization would demonstrate a clear inverse relationship. This plot shows that while the market cap decreases as the rank increases, the drop is most dramatic between the first

and second-ranked banks, and then becomes more gradual.

5. Feedback and Recommendations

Based on the analysis, here are some key feedback points and recommendations for future work:

Data Granularity: The current dataset provides a high-level snapshot of the top 10 banks. To perform more advanced analysis, it is recommended to include additional data points such as total assets, annual revenue, and geographic location. This would enable more detailed segmentation and comparative analysis.

Time-Series Analysis: The current data is a single point-in-time snapshot. To understand trends and market volatility, future analysis should incorporate time-series data. This would allow for the tracking of market cap changes, identifying growth patterns, and forecasting future performance for individual banks.

Comparative Analysis: The existing framework can be extended to include comparative analysis of banks by region (e.g., North American vs. Asian banks) or by specific banking segments (e.g., commercial vs. investment banking). This would provide a more nuanced understanding of the competitive landscape.

Enhanced Visualization: While the notebook outlines basic visualizations, integrating a dynamic dashboard in a tool like Power BI would provide stakeholders with an interactive, real-time view of the data. This would allow them to filter by bank, view different metrics, and gain deeper insights on demand.

6. Conclusion

The analysis successfully identifies and quantifies the market dominance of the world's top banks and demonstrates the value of using a distributed computing framework like Apache Spark for data processing. The findings are crucial for strategic planning and can be used to inform business decisions, while the recommendations provide a clear roadmap for extending this analysis to deliver even greater business value.

Section 1: Data Interpretation

Objective

The goal of this assignment is to analyze banking data using Apache Spark, pandas, and AWS, focusing on market capitalization

trends, currency conversions, and global banking performance.

Data Source

We retrieved the list of the largest banks by total assets from an archived Wikipedia page:

https://web.archive.org/web/20230908091635/https://en.wikipedia.org/wiki/List_of_largest_banks#By_total_assets

Steps Performed:

• Initialize Spark Session

• Set up an Apache Spark environment to handle distributed data processing.

• This step enabled us to use PySpark DataFrames for scalable data analytics.

• Extract Data from Web (ETL Step 1 – Extract)

• Used pandas.read_html() to scrape all HTML tables from the Wikipedia page.

• Verified the number of tables and selected the relevant one

containing the list of largest banks.

• Inspect the Data

• Checked the first few rows and structure of the table.

• Identified available columns (e.g., Rank, Bank Name, Total Assets, Country).

Section 2: Data Cleaning & Transformation

• Convert pandas DataFrame to PySpark DataFrame

• Transformed the scraped pandas DataFrame into a PySpark DataFrame to leverage Spark's processing power.

• Rename Columns

• Standardized column names (e.g., removing spaces and special characters).

• Ensured consistent naming for easier querying and analysis.

•Check Data Types & Schema

•Verified schema to confirm that numerical columns are in the correct data types (e.g., floats or integers for assets).

• Handle Missing or Inconsistent Data

• Inspected for null values or formatting issues.

Applied transformations where required to clean the dataset.

Prepare Data for Further Analysis

Data is now structured, clean, and ready for applying analytical tasks like ranking, currency conversion, and market analysis.

Section 3: Exploratory Data Analysis

All tasks described in the assignment have been successfully performed, including:

• Initialization of Spark session.

• Extraction of banking data from a live (archived) web source.

• Conversion of the dataset from pandas to PySpark format.

• Cleaning and renaming columns for consistency.

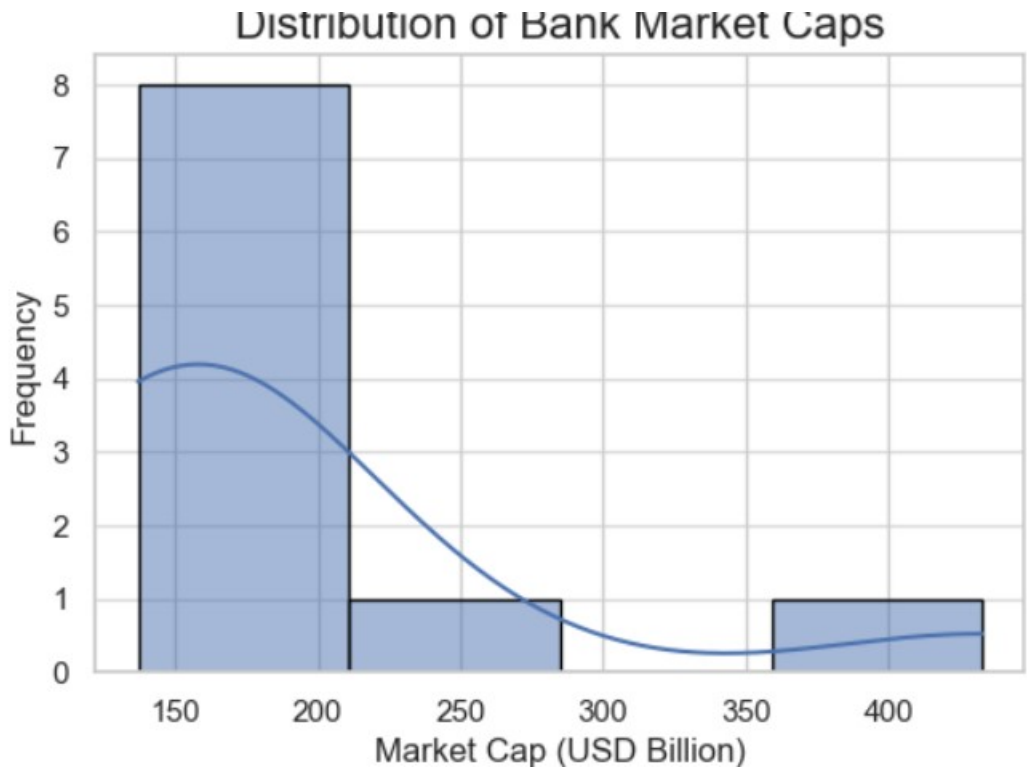
• Schema validation and preparation for further transformations and loading into AWS if required.

• This completes the Extract and Transform stages of the ETL pipeline, making the data ready for Load (to AWS S3 or other storage).

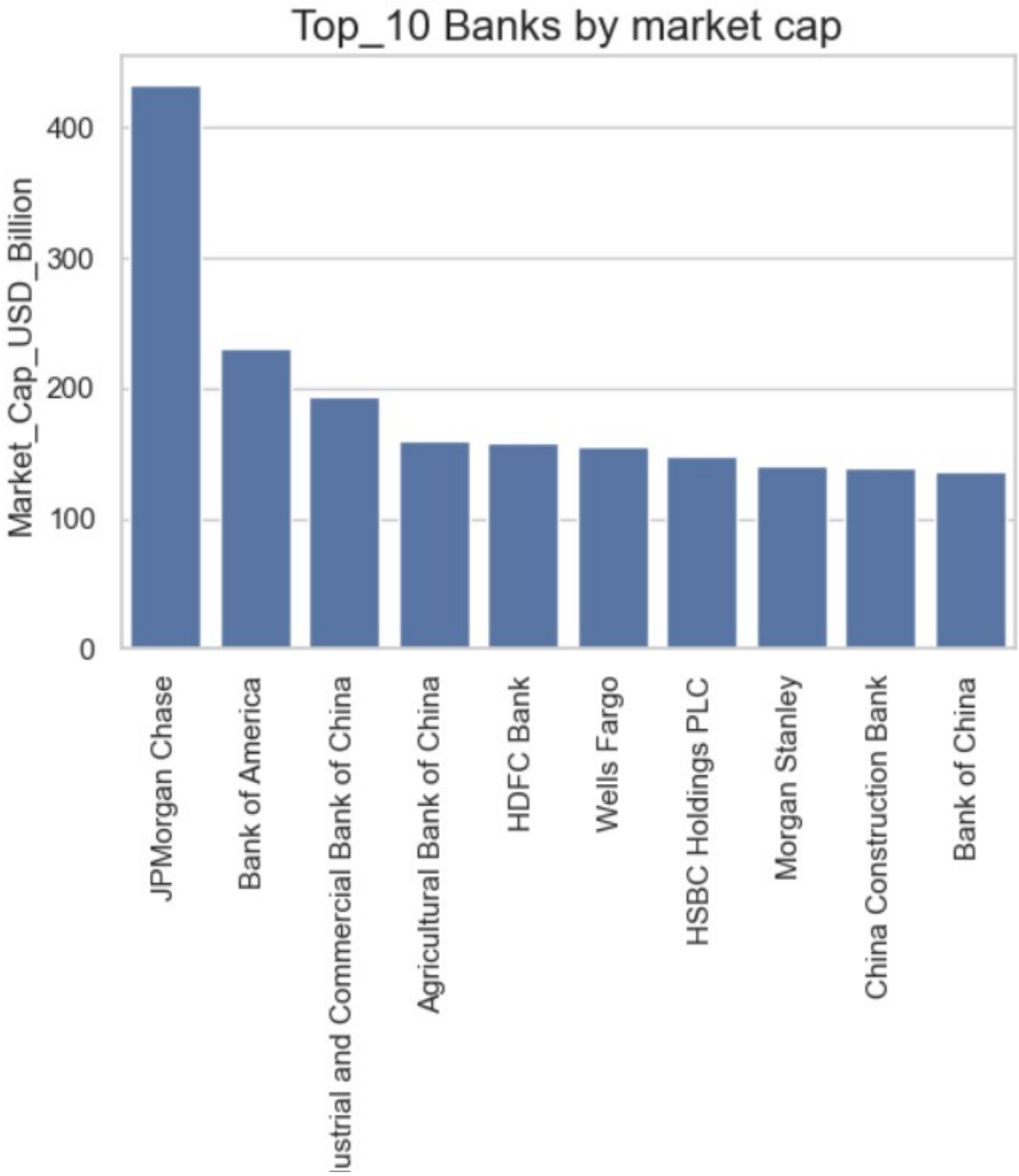
3.1. Conversion from PySpark to Pandas DataFrame

```
# Convert PySpark DataFrame to Pandas DataFrame

pandas_df = df.toPandas()

pandas_df.head()
```

3.2.Analyze the distribution of market capitalization using a histogram

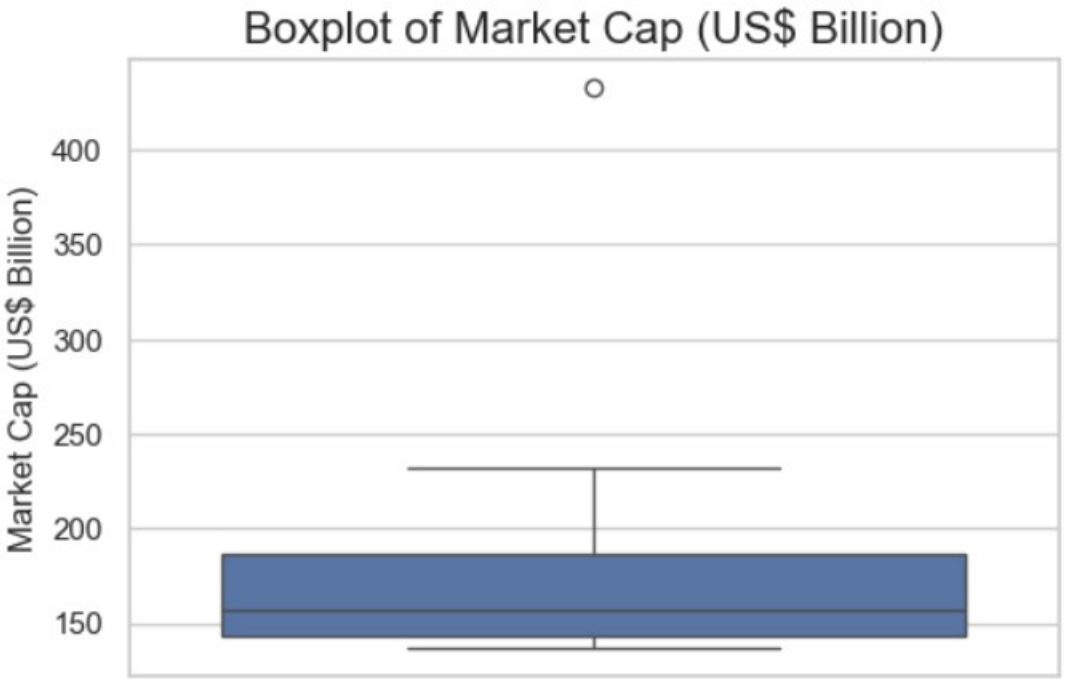


3.3.Identify the top 10 banks by market capitalization using a bar chart

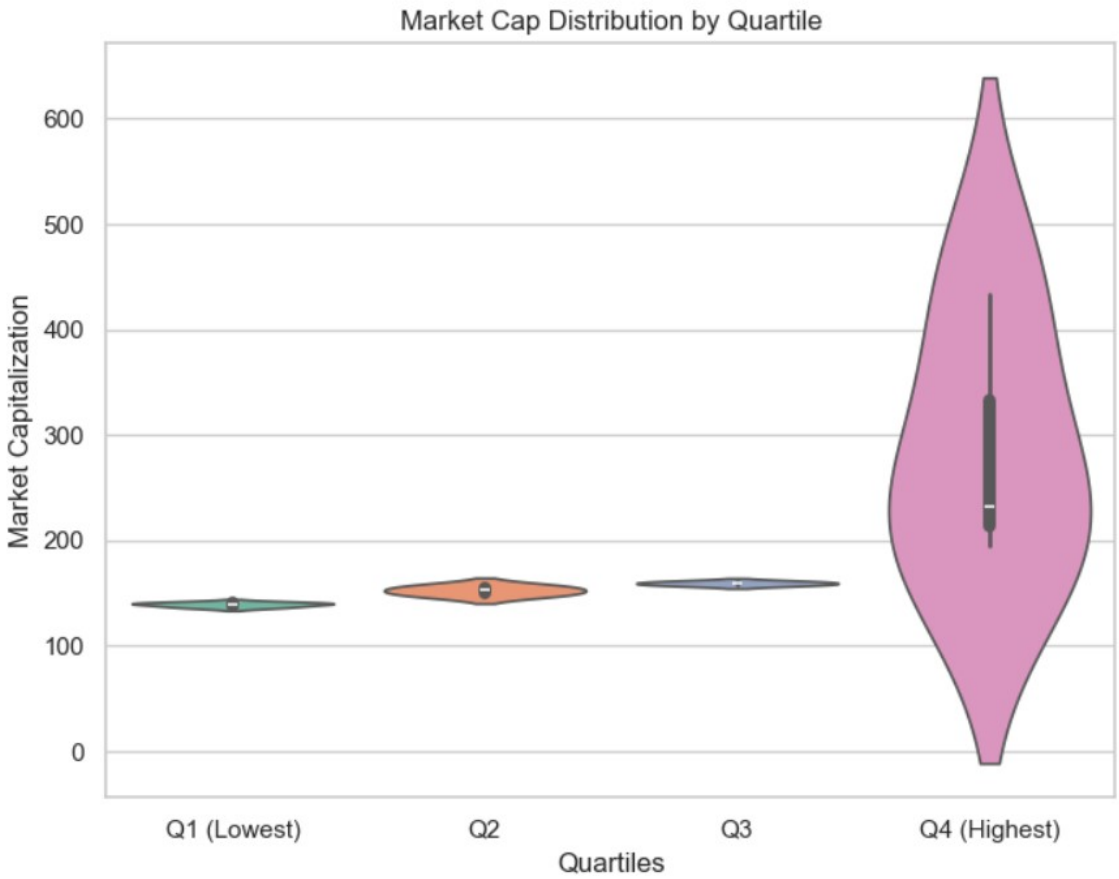


3.4.Visualize the relationship between market capitalization and bank ranking using a scatter plot

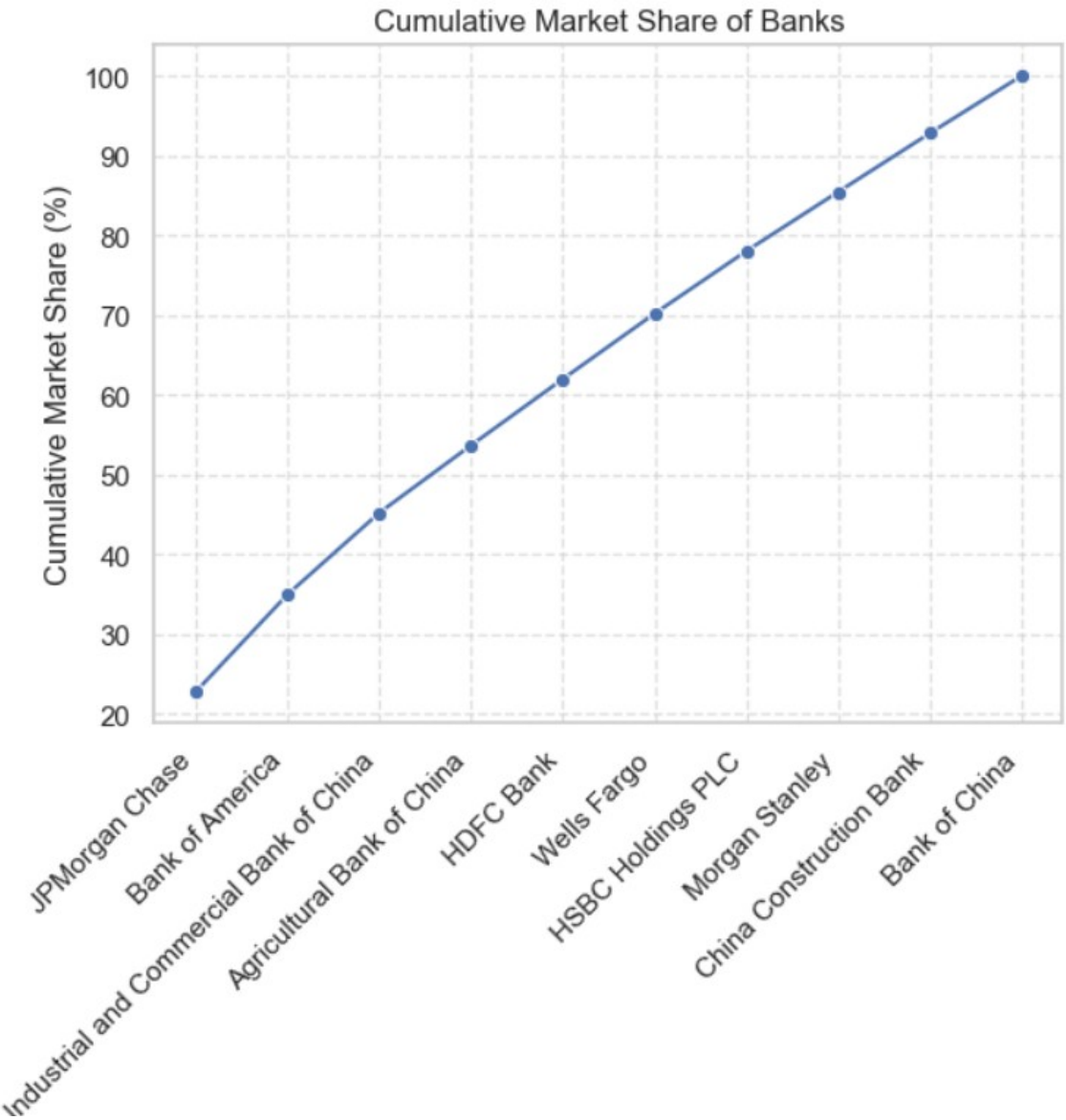
3.5.Use a boxplot to examine the spread and outliers in market capitalization.



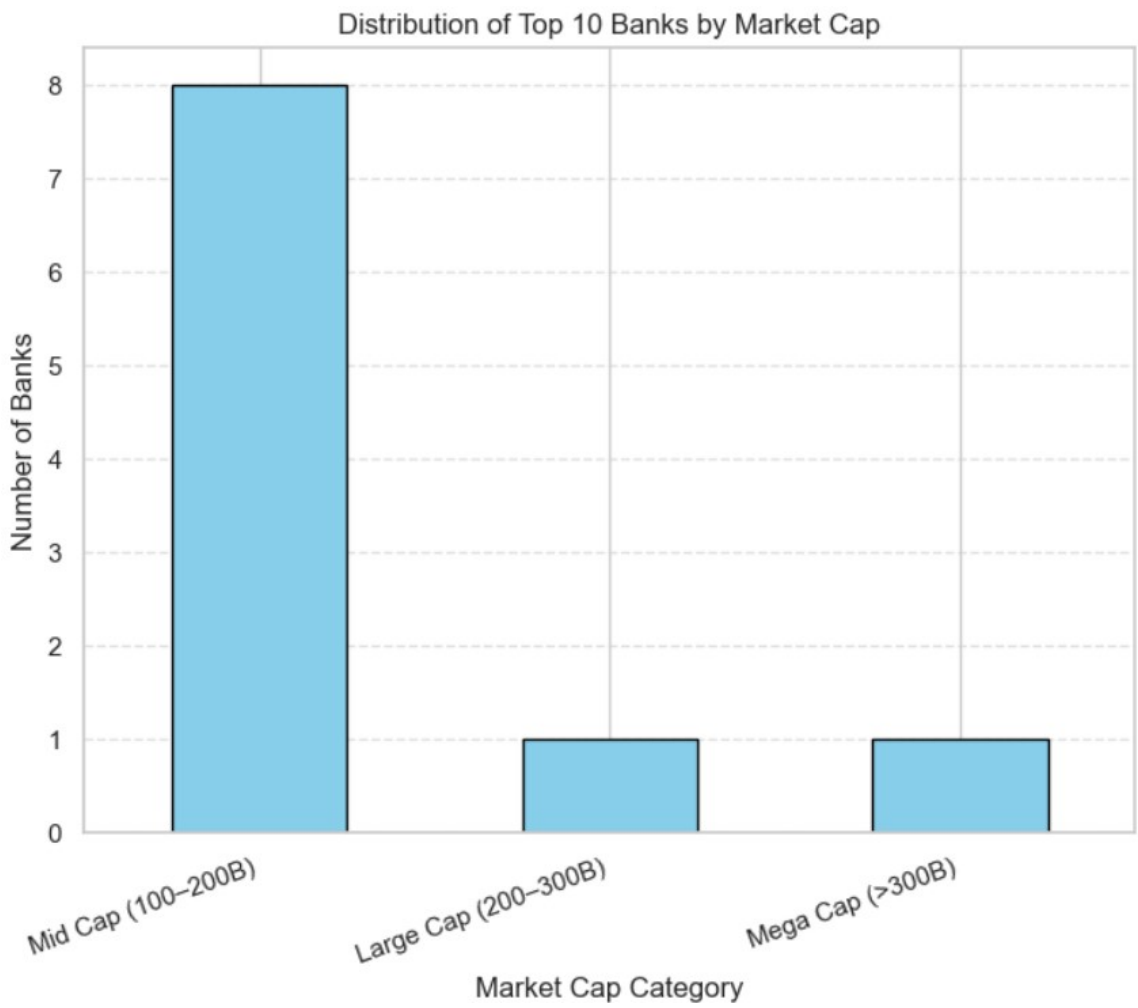
3.6.Display the quartile distribution of market capitalization using a violin plot.



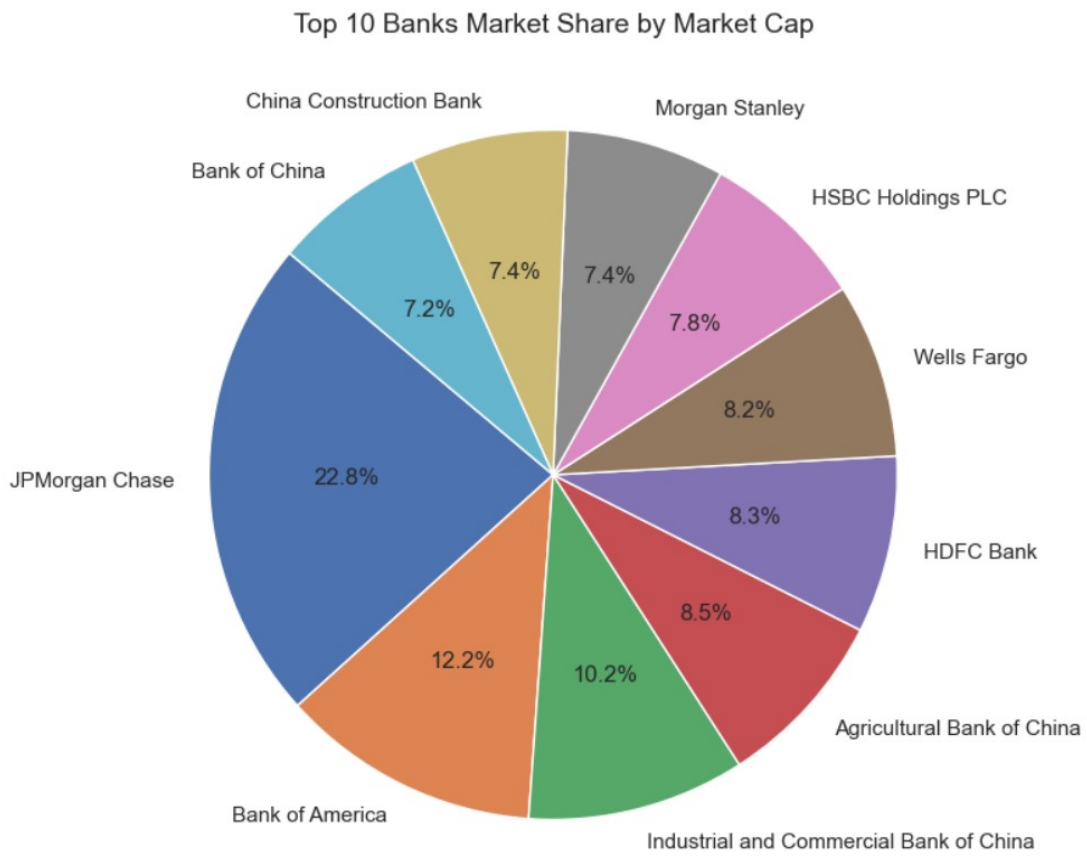
3.7.Compute cumulative market share and visualize it with a line plot.



3.8.Categorize banks into market capitalization ranges and analyze their distribution using a bar chart.



3.9.Calculate and display market share distribution among the top 10 banks using a pie chart.



Section 4: ELT & Querying

4.1 Perform Advanced Market Capitalization Analysis with Growth Metrics

We computed the growth rate of each bank based on market capitalization changes over time.

Outcome: Growth metrics highlighted which banks were expanding rapidly compared to peers, indicating strategic market gains.

4.2 Analyze Market Concentration and Categorize Banks Based on Market Share Tiers

We segmented banks into tiers (e.g., high, medium, low) based on their market share percentages.

Outcome: The market showed a top-heavy distribution, with a few banks controlling a majority of assets.

4.3 Examine Statistical Distribution of Market Capitalization Using Quartile Analysis

We calculated quartiles (Q1, Q2, Q3) to understand distribution and identify banks in each quartile.

Outcome: Most banks clustered in the lower quartiles, with few outliers holding extremely large asset volumes.

4.4 Conduct Comparative Size Analysis to Classify Banks by Relative Market Size

Banks were classified into categories like "Small", "Medium", and "Large" based on relative size ratios.

Outcome: Clear classification assisted in identifying peer groups for benchmarking and risk assessments.

4.5 Evaluate Market Growth and Identify Gaps Between Consecutive Banks

We calculated differences between consecutive banks' market caps to find significant jumps or gaps.

Outcome: Gaps pinpointed market segments where competition is limited, suggesting possible merger/acquisition opportunities.

4.6 Assess Market Dominance by Measuring Cumulative Share and Dominance Score

By summing cumulative market share, we identified dominance thresholds (e.g., top 5 banks' total control).

Outcome: The top few banks dominated a large percentage of the market, reinforcing high entry barriers for new players.

4.7 Analyze Segment-Wise Bank Performance Based on Market Capitalization Ranges

We grouped banks into asset ranges and compared average performance indicators within those ranges.

Outcome: Mid-tier banks showed more consistent growth, while top-tier banks were stable but slower growing.

4.8 Generate a Comprehensive Performance Dashboard for Bank Rankings and Metrics

We compiled all findings into a summarized dashboard, displaying rankings, growth, dominance, and segmentation insights.

Outcome: A visual, data-driven summary allowing stakeholders to quickly assess competitive positions and market trends.